

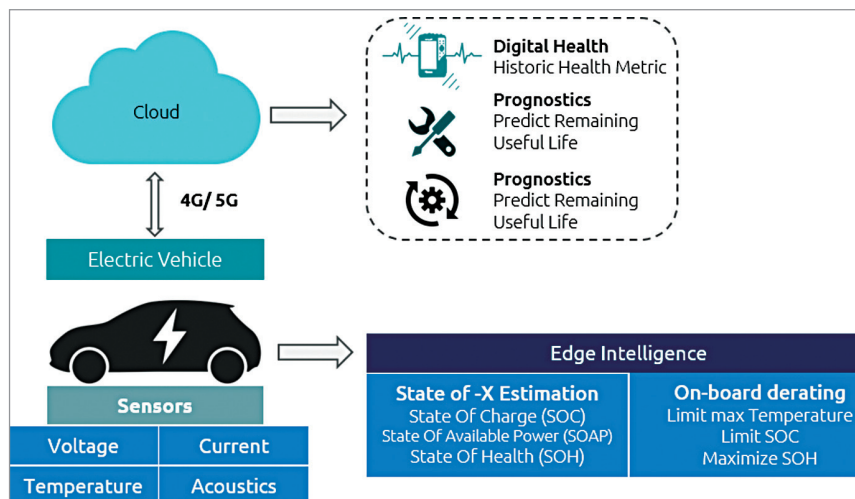


Fig. 4: A typical integration of BMS with IoT and AI

used, which reduces dependence on the cloud.

Most EVs have embedded computing systems that work in real time to do the basic operations automatically, such as the State of X-Estimation—where X stands for State Of Charge (SOC), State of Available Power (SOAP), and the State Of Health (SOH).

The actions taken by edge intelligence can help limit certain anomalies such as the maximum temperature by lowering the power consumption by the motor. Although this can make the EV slightly slower, the motor won't draw excessive power and overheat the battery. Edge intelligence can also control the State of Charge to maximise the State Of Health.

With the help of 4G/5G, the data can be sent to the cloud. Graphic processing units (GPUs) with huge computing power can calculate digital health and predict the remaining useful life on the cloud. If an anomaly occurs in the BMS, a notification is sent to the registered mobile phone or on any dashboard application installed in the EV.

Basically, a digital twin of an EV and every part of it (such as a battery, motor, controller) is created. And data from different sensors are analysed to deliver the safety of the driver, passenger, and the vehicle itself.

Some benefits of integrating BMS with IoT are:

- A mirror framework (digital twin) of processes and functions of a car is created on the cloud or edge

- Sends updates and maintenance notifications from the car to the maintenance engineers, or the car owner/company, so that they can guide for any servicing requirements

- Initiates system warning in case of any anomaly

- Management of energy stored in a battery and the amount of charge required (smart charging)

- SOX estimation
- Battery diagnostics
- Cell balancing
- Data collection and processing

Big car companies like Tesla and MG have already installed different kinds of sensors in their cars. So, on swapping a battery, you receive a notification that the battery has been uninstalled/installed in a vehicle. Also, these batteries come with an electronic control unit (ECU), which has its digital properties and identity.

On the topic of data privacy, it does not matter which company provides the battery. Generally, the data-sharing policies of a car manufacturer play a decisive role. When a car company shares data with the battery manufacturer, the latter can optimise battery production according to the requirement and save a lot of time and cost. **EFY**

*The article is based on the talk 'The Combination of BMS and IoT In EVs' by Arqum Ahmad, Scientist (Digital Twin) at Leonardo S.p.A, Italy, presented at February edition of Tech World Congress 2021. He is also the co-founder and former CEO of Antrax Labs, India. The article is compiled by Vinay Prabhakar Minj, a technology journalist at EFY*

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